3.8 Linear Equation in Two Variables

A linear equation in two variables is of the form $ax + by + c = 0$ where a, b and c are real numbers, both a and b are not zero (The two variables are denoted here by x and y and c is a constant).

Examples

Linear equation in two variables	Not a linear equation in two variables.
$2x + y = 4$	$xy + 2x = 5$ (Why?)
$-5x + \frac{1}{2} = y$	$\sqrt{x} + \sqrt{y} = 25$ (Why?)
$5x = 35y$	$x(x+1) = y$ (Why?)

If an equation has two variables each of which is in first degree such that the variables are not multiplied with each other, then it is a linear equation in two variables (If the degree of an equation in two variables is 1, then it is called a linear equation in two variables).

An understanding of linear equation in two variables will be easy if it is done along with a geometrical visualization (through graphs). We will make use of this resource.

Why do we classify, for example, the equation $2x + y = 4$ is a linear equation? You are right; because its graph will be a line. Shall we check it up?

We try to draw its graph. To draw the graph of $2x + y = 4$, we need some points on the line so that we can join them. (These are the ordered pairs satisfying the equation).

To prepare table giving ordered pairs for $2x + y = 4$. It is better, to take it as

$y = 4 - 2x$. (Why? How?)

When $x = -4$, $y = 4 - 2(-4) = 4 + 8 = 12$

When $x = -2$, $y = 4 - 2(-2) = 4 + 4 = 8$

When $x = 0$, $y = 4 - 2(0) = 4 + 0 = 4$

When $x = +1$, $y = 4 - 2(+1) = 4 - 2 = 2$

When $x = +3$, $y = 4 - 2(+3) = 4 - 6 = -2$

Thus the values are tabulated as follows:

x-value	-4	-2	0	1	3
y-value	12	8	4	2	-2

(To fix a line, do we need so many points? It is enough if we have two and probably one more for verification.)

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